

"Agentic AI Alpha Assignment"

day / date: 05/10/2026

Ans 1) Pixels = $1280 \times 720 = 921,600$

$$x = \frac{29,491,200}{921,600} = 32 \text{ bits}$$

2) $\frac{\text{Bytes}}{\text{Frame}} = \frac{29,491,200}{8} = 3,686,400 \text{ bytes}$

3) $\frac{\text{Bits}}{8} = 70 \times 10^6 \times 8 = 560 \times 10^6 \text{ bits}$

Ans 2) $\text{Score} = \frac{(Q \cdot K)}{\sqrt{d}}$, $d = \sqrt{2} = 1.414$

Query AI	AI	Grov	Emerging	Course
	8.25	10.5	10.5	10.65
Governance	11	14	14	14.2
Emerging course	12.1	15.4	15.4	15.62
	12.32	15.6	15.6	15.84

For AI:-

Scaled: 5.83, 7.43, 7.43, 7.53

Softmax weights: 0.061, 0.302, 0.202, 0.335

Output: [2.02, 0.30]

~~Ans 2~~ For Governance:-

Softmax weights: 0.037, 0.306, 0.306, 0.352

Output: [2.07, 0.30]

For emerging:-

Softmax weights: 0.030, 0.306, 0.306, 0.358

Output: [2.08, 0.30]

~~Ans 2) 0.30, 0.30, 0.30, 0.30~~

For course:-

Softmax weights: 0.030, 0.305, 0.305, 0.361

Output: [2.08, 0.30]

Ans 3)

Naive Bayes (Laplace):-

Score = prior $\times$ P(wind) $\times$ P(humidity)

Wind, Humidity	Yes	No	Rain	Decision
Weak, High	0.216	0.050	81%	Hold
Strong, High	0.144	0.050	74%	Hold
Weak, Low	0.144	0.150	49%	Deliver
Strong, Low	0.096	0.150	39%	Deliver

Entropy & Information gain:-

$$H(S) = 0.971$$

$$IG(\text{Wind}) = 0.020$$

$$IG(\text{Humidity}) = 0.420$$

High gives rain

Low splits on Wind: sometimes no, sometimes tie

Logistic Regression:-



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Fit: $\text{logit} = -0.04 - 0.95 \cdot \text{strong} + 2.53 \cdot \text{High}$, L2 regulariser, $C=10$. No regularisation $\rightarrow$ weights diverge.

$P(\text{rain}) = 0.92$ (Weak, High)

$P(\text{rain}) = 0.82$ (Strong, High)

$P(\text{rain}) = 0.049$ (Weak, Low)

$P(\text{rain}) = 0.27$ (Strong, Low)

Comparison:-

- Same decision for every case except Weak/Low
- Weak/Low is ambiguous, training data is equally distributed.
- Tree is interpretable, but ties on this case
- Logistic regression gives probabilities, but needs regularisation.

Ans 5)

(1) $P(A \cap B) = 0.5 \times 0.5 = 0.25$

(2) Sprayed Fields turn healthy and can't be re-infected before $t=10$.

t	Location	Percept	Action	Credit
0	A	Infected	Spray $\rightarrow$ B	\$0
1	B	Infected	Spray $\rightarrow$ A	\$0
2	A	Healthy	no $\rightarrow$ B	\$2
3	B	Healthy	$\rightarrow$ A	\$2

Total = \$4



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Ans 7)

$$1) A \cdot b = 6(4) + (-1)(x) + 3(-2) = 24 - x - 6 = 0$$

$$x = 18$$

2) Dot Product multiplies matching entries and sums them, giving a scalar. Hadamard product multiplies matching entries and keeps them separate, giving a vector.

$$3) |v| = \sqrt{8} = 2\sqrt{2}$$

$$u \cdot \text{proj} = (u \cdot v / |v|^2) \cdot v = (6/8)(2\sqrt{2}, 2) = 1.5\sqrt{2}$$

$$u \cdot v = 1.5\sqrt{2} \times 2\sqrt{2} = 6$$

$$\text{Ans 8) } S(t) = 0.2t + t^2 + [\sin t + \cos t] + [t^5 + t^6 + \sin t] + [t^5 + t^8 + \sin t + \cos t]$$

t	A	B	C	D	S(t)
1	1.2	1.01	2.01	3.0173	7.252
2	4.4	1.03	96.0	289.03	390.503
3	9.6	1.05	972.0	6805.05	7787.754
4	16.8	1.06	5126.0	66561.06	71699.004

Ans 6)

1) Models agent interactions as games with strategies and payoffs, so we can predict cooperation or competition and compare agents' actual behaviour with theoretical equilibria to spot bias.

2) It makes multi-agent outcomes predictable and interpretable by showing how



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individual incentives conflict with collective benefit. That supports trustworthy, fair AI.

- 3) 1) Prisoner's Dilemma: cooperate or defect.
Mutual defection is the Nash equilibrium, even though mutual cooperation pays more.
Battle of the sexes: a coordination game where players prefer different outcomes but both want to match.

- 4). Llama 3.1 405B
• Mistral Large
• GPT-4o
• Claude 3.5 Sonnet

1) Dec-POMDP:-

$(N, S, A, P, R, \Omega, O, n, \gamma)$

- $N \rightarrow$ no. of agents
- $S \rightarrow$ global env states
- $A \rightarrow$ joint actions of all agents,
- $P \rightarrow$ state transition probability
- $R \rightarrow$ reward function
- $\Omega \rightarrow$ possible observations
- $O \rightarrow$ observable function
- $\gamma \rightarrow$ discount factor

2) • Hard Sharing: some layers/parameters are shared, while task/agent-specific layers remain separate. It is used to reduce computation / storage and speed up learning, while maintaining specialization

Soft sharing: Each task has its own



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parameters, but the models are encouraged to remain similar.

$$3) L_e = E \left[\sum_i (\|\hat{o}_i - o_i\|^2 + \lambda_e \|\hat{r}_i - r_i\|^2) \right]$$

The model tries to predict the next observation (o_i) and the reward r_i . λ_e controls the importance of reward prediction. Goal is to minimize the loss.

$$4) r_t^{\text{tot}} = r_t + \lambda_e r_t^e + \lambda_d r_t^d$$

- $r_t \rightarrow$ normal reward
- $r_t^e \rightarrow$ encourage exploration
- $r_t^d \rightarrow$ encourages different agent types to have distinguishable representations.

The agent is not only rewarded for task completion, but also encouraged to explore new situations and learn representations that distinguish different agent types.

